

Rishi Singh

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EDUCATION

University of British Columbia

Bachelor of Applied Science - Electrical Engineering (Co-op program)

Available May 2026 for 4-16 months

Vancouver, BC

Expected Graduation: April 2028

TECHNICAL SKILLS

Languages: C/C++, Verilog/SystemVerilog (HDL), Python, Assembly (ARM/8051), Bash/Shell scripting

Developer Tools: Quartus, ModelSim, Vivado, Altium Designer, Git, MATLAB, LTSpice, Solidworks, Linux

Lab Equipment: FPGA (DE1-SoC), Oscilloscope, Digital Multimeter, Waveform Generator, SMT/THT Soldering, Reflow Oven

EXPERIENCE

Undergraduate Research Assistant

UBC Sensorimotor Physiology Laboratory

Jan 2026 – Present

Vancouver, BC

- Built a real-time data acquisition system on a Raspberry Pi 5 synchronizing 6-axis IMU data (up to 1660 Hz via I2C) with 120fps video from a TechNexion AR0234 global shutter camera, using a GStreamer H.264 pipeline with PTS-based frame timestamps
- Benchmarked and optimized video encoding, reducing frame drop rate from 25.3% to 14.1% at 720p/120fps; authored a migration proposal to NVIDIA Jetson Orin Nano for hardware-accelerated encoding to achieve zero frame drops

Hardware and Firmware Co-Lead

SUBC, UBC's submarine design team

Sep. 2024 – Present

Vancouver, BC

- Designed and fabricated a 6-layer PCB (50×50mm) in Altium Designer integrating nRF52840 SoC, environmental sensors, and multi-protocol communication (SPI, I2C, UART, LVDS)
- Trained 15 members in Altium Designer, SMT/reflow soldering, and component sourcing; initiated an electronic steering system project, coordinating research across motor drivers and feedback sensors

Avionics Systems Technician

Royal Canadian Air Force (436 Squadron)

Sep. 2018 – Sep. 2023

Trenton, ON

- Executed 24/7 maintenance, testing, and calibration on critical avionics systems (Radar, Crypto/Communication, Flight Control, Autopilot, TCAS) for CC-130J Hercules aircraft
- Ensured high operational readiness supporting key global missions, including the evacuation of 3,700 civilians (Op AEGIS) and the delivery of 16 million pounds of military aid (Op REASSURANCE)

PROJECTS

ARC4 Parallel Cracker | *SystemVerilog, Quartus, Cyclone V FPGA*

Nov. 2025

- Designed a complete ARC4 encryption/decryption core using hierarchical FSMs (Init, KSA, PRGA) with ready-enable handshake protocols, interfacing custom RTL with 209 M10K embedded memory blocks
- Scaled to 104 parallel cracking cores with PLL clock boosting and early-termination ASCII validation, exhausting the full 2^{24} key space in 1.56s (10.75M keys/sec) on a Cyclone V FPGA — a $94\times$ speedup over single-core baseline

Data Acquisition Module PCB ([link](#)) | *Altium Designer, C, Zephyr RTOS*

Aug. 2025

- Designed 6-layer stackup optimizing antenna/SD card isolation; integrated nRF52840 MCU with SPI SD card logging, I2C/UART sensors, and LVDS transceivers for submarine sensor systems
- Fabricated and currently debugging board bring-up, troubleshooting power delivery and flash programming to prepare for Zephyr RTOS firmware development on nRF52840 SoC

VGA Graphics Engine | *SystemVerilog, DE1-SoC FPGA*

Oct. 2025

- Designed FSM-driven VGA controller rendering Bresenham's Circle (8-octant symmetry) and Reuleaux Triangle using fixed-point $\sqrt{3}$ arithmetic on a 160×120 pixel display at 25MHz pixel clock

Reflow Oven Controller | *8051 Assembly, Analog Circuit Design*

Feb. 2025

- Implemented a 5-stage reflow profile FSM in 8051 assembly; designed analog thermocouple conditioning circuit achieving $\pm 1.7^\circ\text{C}$ accuracy across 15–260°C range